

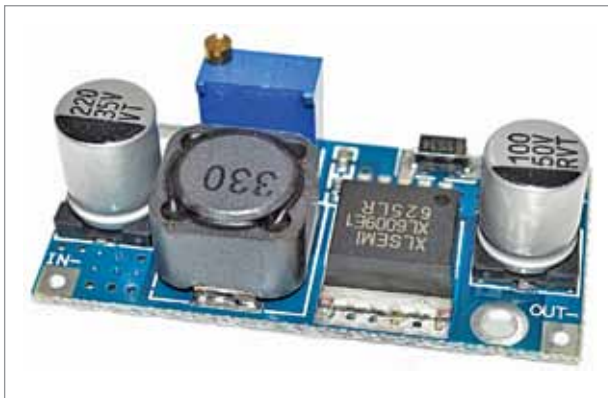


Fig. 4: XL6009 module

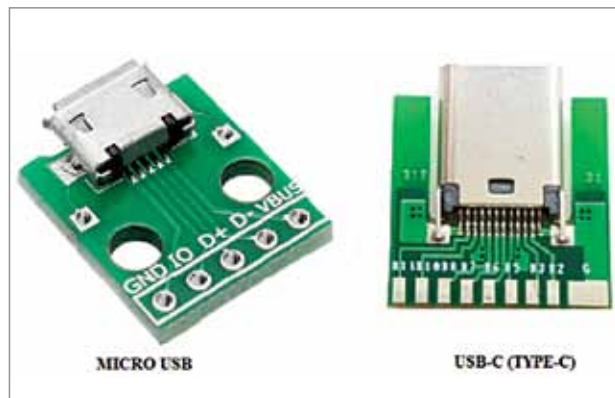


Fig. 5: Types of USB breakout boards

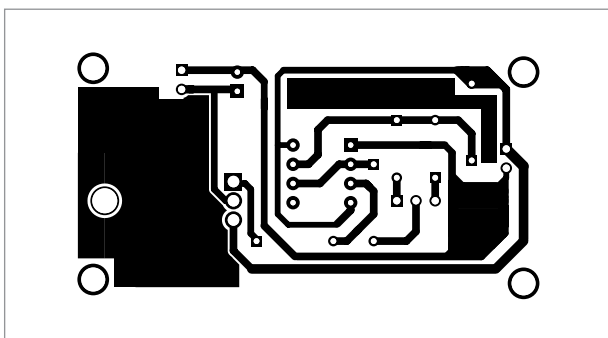


Fig. 6: Actual-size PCB layout of COB LED table lamp

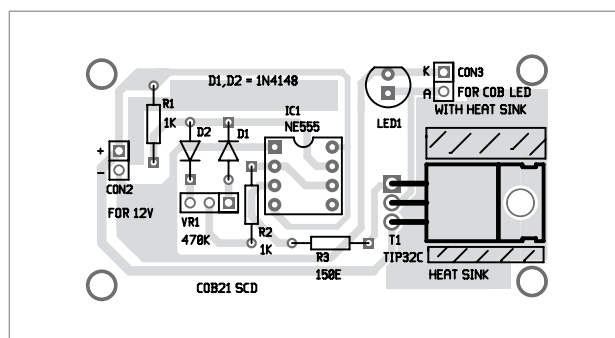


Fig. 7: Components layout of the PCB

PARTS LIST

Semiconductors:

IC1	- NE555 timer
T1	- TIP32C PNP transistor
LED1	- 20W COB LED
D1-D2	- 1N4148 signal diode
Board1	- XL9006 DC-DC booster/ converter

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2	- 1-kilo-ohm
R3	- 150-ohm
VR1	- 470-kilo-ohm potmeter

Capacitors:

C1	- 0.1µF ceramic
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Miscellaneous:

CON1	- Micro USB or Type-C (USB-C) connector
CON2, CON3	- 2-pin connector - Heat-sink for T1 - 8-pin socket for IC1 - Heat-sink for LED1 - Old LED table lamp

TIP32C power transistor is a 3-pin TO-220 plastic-package PNP transistor. It is used to drive the COB LED. It can drive a collector current up to 3A and has a maximum junction temperature up to 150°C. It is recommended to use a suitable heat-sink with it.

XL6009 booster module. This DC-DC converter/booster module

can accept wide input range from 3V to 32V and produce output from 5V to 35V. It has an inbuilt potentiometer to control and adjust the output voltage. This module is a non-isolated step-up (boost) voltage converter featuring adjustable output voltage and high efficiency. The IN+ and IN- are inputs for positive and negative, OUT+ and OUT- are outputs for positive and negative, respectively. The XL6009 module used in the project is shown in Fig. 4.

The main advantage of this circuit is that you can connect it with any USB 5V DC mobile charger. You can connect a 5V DC to input terminals (IN+ and IN-) of XL6009 module through micro USB or type-C USB connector. Many breakout boards for micro and Type-C USB are available in the market (Fig. 5). Depending on your charger, you can use any of these breakout boards.

With an input of 5V at CON1, adjust the output across OUT+ and OUT- terminals to 12V using the inbuilt potmeter of XL6009 module and a voltmeter. Then connect the OUT+ and OUT- terminals to CON2

connector of the circuit.

Working of the circuit is straightforward. After connecting the 12V DC supply to CON2, COB LED will glow. Vary VR1 to adjust the brightness.

Construction and testing

You can assemble the circuit on a general-purpose PCB or a custom designed PCB shown in Fig. 6 and Fig. 7. Use an 8-pin socket for IC1 and suitable heat-sink for power transistor T1. After mounting the components on the PCB, enclose the assembly in a suitable box.

Open the base cover of your existing old table lamp. Remove the old inbuilt circuit and replace it with your circuit. Carefully remove the old LED and replace it with the COB LED. Connect the circuit to power supply using any mobile charger or 5V DC power supply.

This COB LED lamp is suitable not only as a table lamp, it gives sufficient light for a small bedroom. **EFY**

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